

GDSC2 · Sanger

967 lines · 285 drugs

CTRP v2.2 · Broad

645 lines · 479 cpds

CCLE · Broad

57,820 genes · 8 omics

Harmonize

cell → DepMap ID · drug → InChIKey (structure) · metric = sens_z (– = sensitive)

Model-ready dataset

810 cell lines · 37 diseases · 678 drugs · log1p TPM + 8 multi-omic layers

Evaluation

grouped leave-cell-line-out nested CV · scored vs drug-mean and tissue baselines

Broad-spectrumdrug-mean ranking
(within-tissue; high-confidence)**Mechanism-specific**tissue-free: disease×drug residualized
(permutation-gated; vs tissue baseline)**Pathway analysis**

ssGSEA pathway activity · per-drug in-fold signatures + permutation null · GSEA atlas

Tiered drug recommendation

broad-spectrum (high-confidence, within-tissue) + mechanism-specific (permutation-validated minority)